

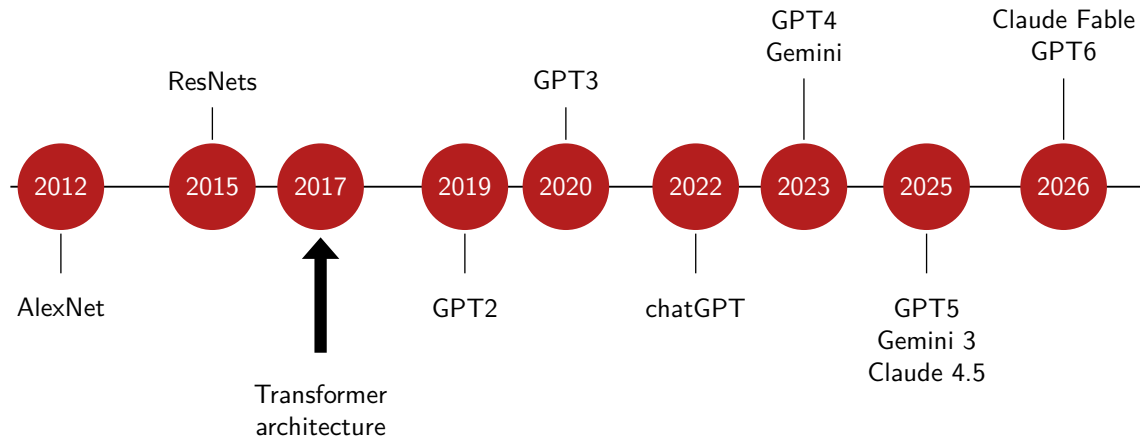
The Mean-Field Dynamics of Transformers

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主讲人：曹溪能

P. Rigollet, arXiv:2512.01868 (2025).

Neural Networks since AlexNet



Attention Is All You Need

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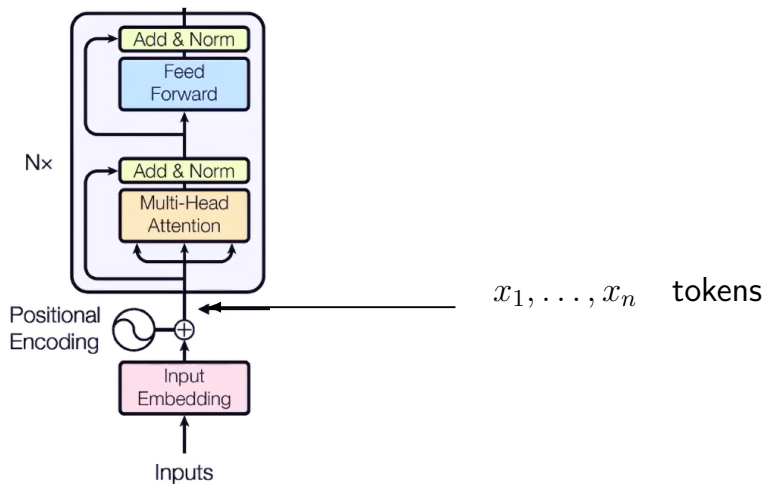
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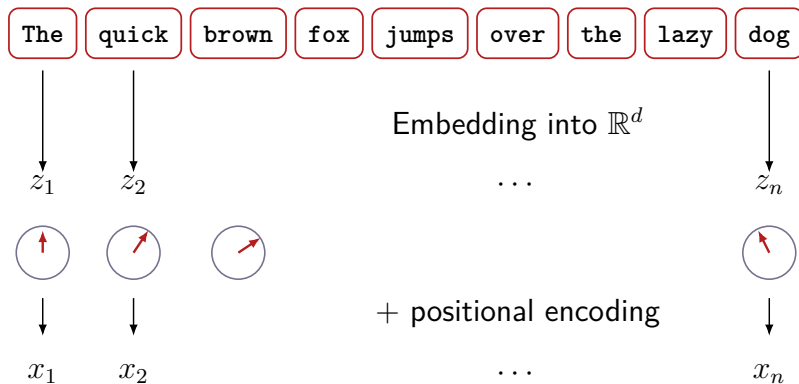
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[Vaswani et al. '17]

Transformers

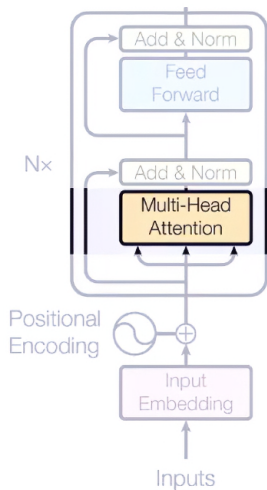


Tokens



$x_1 \dots x_n \in \mathbb{R}^d$: **tokens**

Transformers

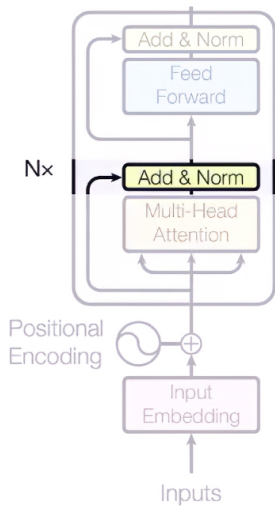


Interacting particle system

$$x_i \leftarrow x_i + \sum_{j=1}^n \frac{e^{\beta \langle Q_{\ell} x_i, K_{\ell} x_j \rangle}}{\sum_{k=1}^n e^{\beta \langle Q_{\ell} x_i, K_{\ell} x_k \rangle}} V_{\ell} x_j$$

Attention head

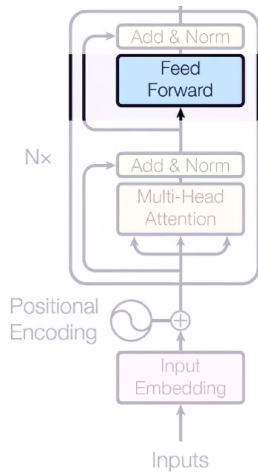
Transformers



$$x_i \leftarrow \frac{x_i}{\|x_i\|}$$

Layer Normalization

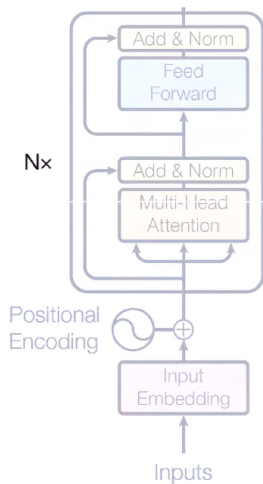
Transformers



$$x_i \leftarrow x_i + \sigma(A_\ell x_i + b_\ell)$$

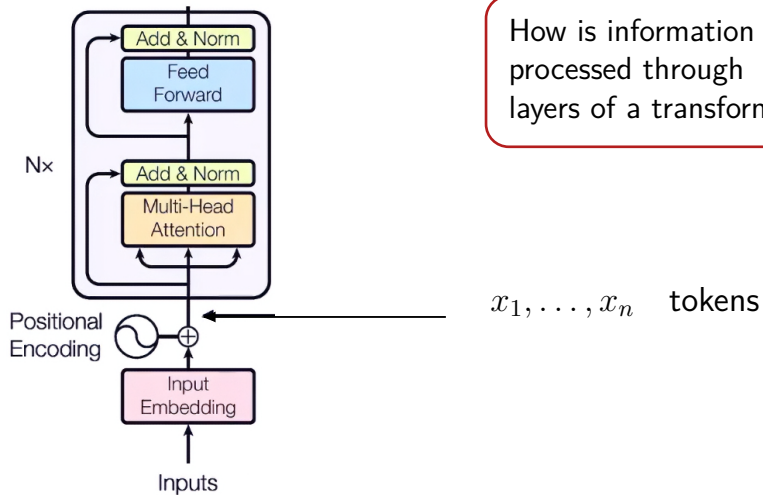
Feed Forward (MLP) layer

Transformers



Repeat for N layers / blocs

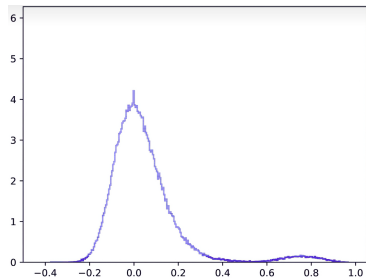
Transformers



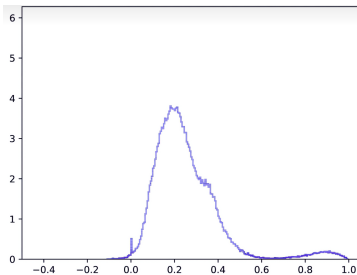
How is information processed through layers of a transformer ?

Pre-trained ALBERT — Histogram of $\{\langle x_i, x_j \rangle\}_{i \neq j}$

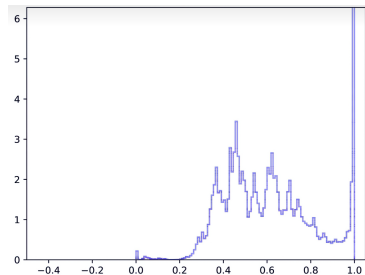
Layer 0



Layer 5



Layer 48



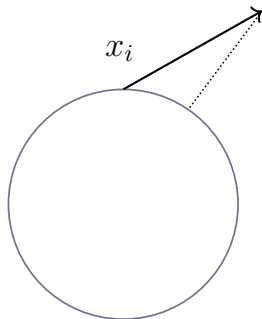
Self-Attention Dynamics

Attention head

$$x_i \leftarrow x_i + \sum_{j=1}^n \frac{e^{\beta \langle Q_\ell x_i, K_\ell x_j \rangle}}{\sum_{k=1}^n e^{\beta \langle Q_\ell x_i, K_\ell x_k \rangle}} V_\ell x_j$$

Layer Normalization

$$x_i \leftarrow \frac{x_i}{\|x_i\|}$$



Self-Attention Dynamics

Attention head

$$x_i \longleftarrow x_i + \sum_{j=1}^n \frac{e^{\beta \langle \cancel{Q_\ell} x_i, \cancel{K_\ell} x_j \rangle}}{\sum_{k=1}^n e^{\beta \langle \cancel{Q_\ell} x_i, \cancel{K_\ell} x_k \rangle}} \cancel{V_\ell} x_j$$

Layer Normalization

$$x_i \longleftarrow \frac{x_i}{\|x_i\|}$$

$$\dot{x}_i(t) = \text{Proj}_{\mathcal{T}_{x_i(t)} S^{d-1}} \frac{\sum_{j=1}^n x_j(t) e^{\beta \langle x_i(t), x_j(t) \rangle}}{\sum_{k=1}^n e^{\beta \langle x_i(t), x_k(t) \rangle}}$$

Mean-field interacting particle system

Mean-Field

$$\dot{x}_i(t) = \text{Proj}_{\mathcal{T}_{x_i(t)} S^{d-1}} \frac{\sum_{j=1}^n x_j(t) e^{\beta \langle x_i(t), x_j(t) \rangle}}{\sum_{k=1}^n e^{\beta \langle x_i(t), x_k(t) \rangle}}$$

$$\dot{x}(t) = \text{Proj}_{\mathcal{T}_{x_i(t)} S^{d-1}} \frac{\int y e^{\beta \langle x(t), y \rangle} \mu(t)(dy)}{\int e^{\beta \langle x(t), y \rangle} \mu(t)(dy)}$$

$$\dot{x}(t) = \frac{1}{\beta} \nabla \log \int e^{\beta \langle x(t), y \rangle} \mu(t)(dy)$$

$$\dot{x}(t) = \frac{1}{\beta} \nabla \log \int e^{-\frac{\beta}{2} \|x(t) - y\|^2} \mu(t)(dy)$$

$$\mu(t) = \frac{1}{n} \sum_{i=1}^n \delta_{x_i(t)}$$

∇ is gradient on S^{d-1}

$$\|x(t)\| = \|y\| = 1$$

Wasserstein Gradient Flow

$$\dot{x}(t) = \frac{1}{\beta} \nabla \log \int e^{-\frac{\beta}{2} \|x(t)-y\|^2} \mu(t)(dy)$$

The (reverse) Wasserstein gradient flow of a functional $\mu \mapsto \mathcal{F}(\mu) \in \mathbb{R}$ is given by

$$\dot{x} = \nabla \frac{\delta \mathcal{F}}{\delta \mu}.$$

If Wasserstein gradient flow, we can hope that $\mu(t) \rightarrow \operatorname{argmax} \mathcal{F}(\mu)$

But $\log \int e^{-\frac{\beta}{2} \|x(t)-y\|^2} \mu(t)(dy)$ is not a functional variation [Sander *et al.* '22]

Wasserstein Gradient Flow

$$\cancel{\log} \int e^{-\frac{\beta}{2}\|x(t)-y\|^2} \mu(t)(dy)$$

$$\dot{x}(t) = \frac{1}{\beta} \nabla \cancel{\log} \int e^{-\frac{\beta}{2}\|x(t)-y\|^2} \mu(t)(dy)$$

$$= \text{Proj}_{\mathcal{T}_{x_i(t)} S^{d-1}} \frac{\int y e^{\beta \langle x(t), y \rangle} \mu(t)(dy)}{\int \cancel{e^{\beta \langle x(t), y \rangle} \mu(t)(dy)}}$$

unnormalized
self-attention

$$\int e^{-\frac{\beta}{2}\|x(t)-y\|^2} \mu(t)(dy) = \delta \mathcal{F}(\mu(t))$$

$$\mathcal{F}(\mu) = \frac{1}{2} \iint e^{-\frac{\beta}{2}\|x-y\|^2} \mu(dx) \mu(dy)$$

Repulsive interaction energy.

Energy Maximization

$$\dot{x}(t) = \nabla \delta \mathcal{F}(\mu)(x(t))$$

$$\mathcal{F}(\mu) = \frac{1}{2} \iint e^{-\frac{\beta}{2} \|x-y\|^2} \mu(\mathrm{d}x) \mu(\mathrm{d}y)$$

But we **maximize** the energy!

Global maximizers: Dirac measures (single cluster)

Does gradient flow get trapped in local max?

2D Dynamics, Kuramoto Oscillators

On the circle, we can parameterize by angles:

$$\dot{\theta}_i(t) = -\frac{\sum_{j=1}^n \sin(\theta_i(t) - \theta_j(t)) e^{\beta \cos(\theta_i(t) - \theta_j(t))}}{\sum_{j=1}^n e^{\beta \cos(\theta_i(t) - \theta_j(t))}} \quad [\text{SA}]$$

$$\dot{\theta}_i(t) = -\frac{1}{n} \sum_{j=1}^n \sin(\theta_i(t) - \theta_j(t)) e^{\beta \cos(\theta_i(t) - \theta_j(t))} \quad [\text{USA}]$$

Kuramoto oscillator:
$$\dot{\theta}_i(t) = -\frac{1}{n} \sum_{j=1}^n \sin(\theta_i(t) - \theta_j(t))$$

In any dimension,

$$\dot{x}(t) = \frac{\sum_{j=1}^n x_j(t) e^{\beta \langle x_i(t), x_j(t) \rangle}}{\sum_{j=1}^n e^{\beta \langle x_i(t), x_j(t) \rangle}} \quad [\text{SA}]$$

$$\dot{x}(t) = \frac{1}{n} \sum_{j=1}^n x_j(t) e^{\beta \langle x_i(t), x_j(t) \rangle} \quad [\text{USA}]$$

Converge to a **single cluster** as $t \rightarrow \infty$ for all $\beta \geq 0$

Rates?

Mean-Field Limit $n \rightarrow \infty$

Mean-field approximation

$$R_0 := \left\| \int_{S^{d-1}} x d\mu_0(x) \right\|_2 > 0.$$

$$\partial_t \mu_t + \operatorname{div}(\mu_t X_t[\mu_t]) = 0$$

$$\mu_0 \ll \operatorname{Unif}(S^{d-1})$$

Thm. (Chen, Lin, Polyanskiy, **R.** 25)

If $\beta < \beta_0(\mu_0)$ then for any $d \geq 2$

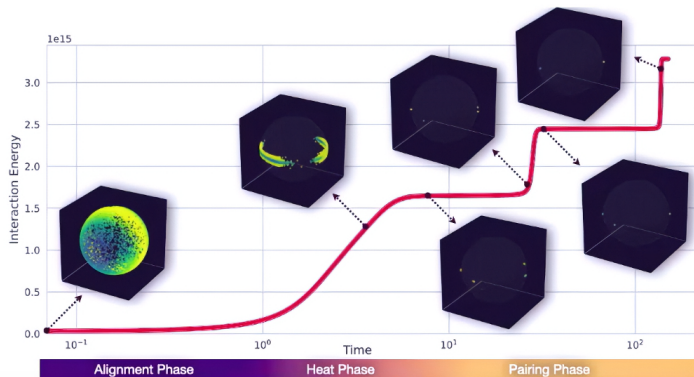
$$W_2(\mu_t, \delta_{x_\infty}) \lesssim e^{-(t-T_0)/99}, \quad \text{for } t > T_0 \simeq d \cdot R_0^{-15}.$$

Mean-Field Limit

See also Bruno, Pasqualotto, Agazzi (NeurIPS '25)

Full dynamical picture for **large** β

$$\partial_t \mu_t + \operatorname{div}(\mu_t X_t[\mu_t]) = 0$$
$$\mu_0 \ll \operatorname{Unif}(S^{d-1})$$



Symmetric Model

Initialize tokens on a simplex:

$$\langle x_i(0), x_j(0) \rangle = \rho \in (0, 1)$$

By symmetry:

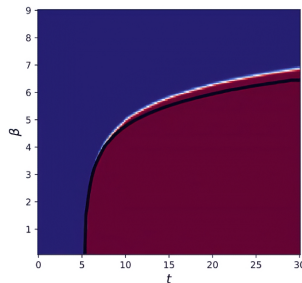
$$\langle x_i(t), x_j(t) \rangle = \rho(t)$$

[Geshkovski *et al.* '25]

[Cowsik *et al.* '24]

If $\rho(0) = 0$, ODE

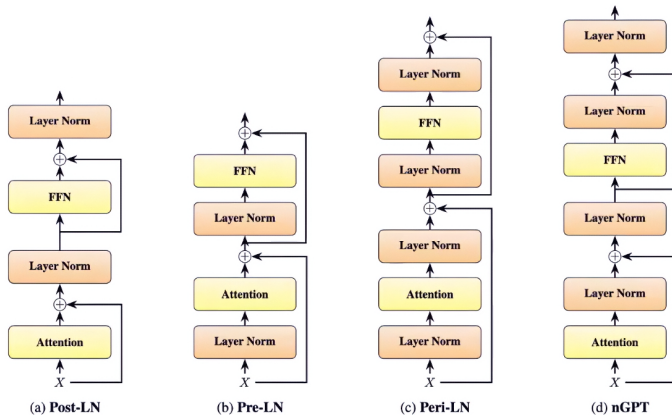
$$\dot{\rho}(t) = \frac{2e^{\beta\rho(t)}(1 - \rho(t))((n - 1)\rho(t) + 1)}{e^{\beta} + (n - 1)e^{\beta\rho(t)}}$$



$n = 32$ points in $d = 512$
dimensions i.i.d. uniform
 $\rho(t) \leq Ce^{-ct}$

Beyond Layernorm

Several normalization schemes



Karagodin, Ge, Polyanskiy, R. NeurIPS '25

Contraction Rates For $\dot{\rho}(t)$

$$t \rightarrow \infty$$

Post-LN

$$C e^{-2t}$$

Pre-LN

$$C/t^3$$

Mix-LN

$$C/t^3$$

Peri-LN

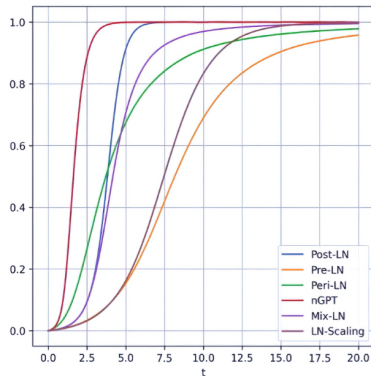
$$C/t^3$$

nGPT

$$C \alpha_t e^{-2 \int_0^t \alpha_s ds}$$

LN-Scaling

$$C \frac{e^{-4\sqrt{t}}}{\sqrt{t}}$$



Fast terminal velocity $\Rightarrow$ representation collapse (useless top layers)

Slow initial velocity $\Rightarrow$ lazy bottom layers

Winner: **PeriLN**

Conclusions

- Simple models for transformers are very rich
- Connections to reverse Wasserstein gradient flows, Kuramoto oscillators, ...
- Multiple heads
- General/random matrices K, Q, V